

Microeconomics

Lecture 11: Profit Maximization

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What We Have Seen So Far

Consumer theory

- Preferences and indifference curves: $MRS = p_x/p_y$ at the optimal choice
- Marshallian demand function: how demand changes as prices and income change
- Income and substitution effects; elasticity

Technology and costs

- Production function, isoquants, MRTS
- $TC(Q) = FC + VC(Q)$; the cost curves: MC , AC , AVC , AFC
- MC intersects AVC and AC at their minima
- Cost minimization: $MRTS = w/r$

Next step

We have studied demand (consumer) and costs (firm) separately. Today we connect them: **how much should a firm produce to maximize profit?**

Learning Objectives

After this lecture you will be able to:

1. **Define** economic profit and distinguish it from accounting profit
2. **Explain** why zero economic profit does not mean earning nothing
3. **Set up** the profit-maximization problem of a *price-taking* firm
4. **Derive** and **interpret** the condition $P = MC$
5. **Check** the second-order condition
6. **Represent** profit graphically as an area
7. **Apply** the *shutdown condition* in the short and long run
8. **Solve** complete numerical exercises with various prices and cost functions

Accounting Profit vs. Economic Profit

Economic profit

$$\pi = TR - TC = P \cdot Q - C(Q)$$

where TC includes **all** opportunity costs: both explicit (monetary payments) and implicit (value of the best alternative use of the resources).

Explicit costs

- Rent, wages, raw materials
- Interest on debt
- Depreciation
- Taxes

Implicit (opportunity) costs

- Entrepreneur's alternative salary
- Alternative return on own capital
- Value of the time employed
- Risk borne

Key difference

Accounting profit includes only explicit costs. **Economic profit** also includes implicit costs. A firm can have positive accounting profit but negative economic profit!

What Does $\pi = 0$ Mean?

Zero economic profit does **not** mean that the entrepreneur earns nothing.

Interpretation of $\pi = 0$

Economic profit is zero when revenues exactly cover **all** opportunity costs, including the entrepreneur's time, capital and risk. The entrepreneur is earning exactly what they would earn in their best alternative: the **normal return**.

Implications:

- $\pi > 0$: the firm is doing **better** than normal $\Rightarrow$ new firms enter
- $\pi = 0$: normal return $\Rightarrow$ no incentive to enter or exit
- $\pi < 0$: the firm is doing **worse** than normal $\Rightarrow$ some firms exit

Example: Accounting vs. Economic Profit

Consider a **pizzeria**. Annual data:

- Total revenue: €140,000
- Explicit costs (ingredients, rent for the premises, employee wages, utilities): €95,000
- Owner's alternative salary as an employee elsewhere: €38,000
- Alternative return on the invested capital (€50,000 at 4%): €2,000

Accounting profit:

$$\pi_{\text{acc}} = 140,000 - 95,000 = 45,000$$

Economic profit:

$$\pi_{\text{econ}} = 140,000 - 95,000 - 38,000 - 2,000 = 5,000$$

Interpretation

The owner earns €5,000 **more** than they would get by closing the pizzeria and working as an employee. The pizzeria is (just) worthwhile. If the alternative salary were €46,000, economic profit would be negative: the owner would be better off closing.

Example: When to Close?

Change of scenario: the following year the market becomes more competitive and revenue falls to €130,000.

Accounting profit:

$$\pi_{\text{acc}} = 130,000 - 95,000 = 35,000 > 0$$

Economic profit:

$$\pi_{\text{econ}} = 130,000 - 95,000 - 38,000 - 2,000 = -5,000 < 0$$

Apparent paradox

Accounting profit is still positive (+€35,000), yet economic profit is negative (-€5,000). What is the best choice? Continue or close?

- The owner gives up €38,000 in salary + €2,000 in return = €40,000 of implicit costs
- They earn “only” €35,000 of accounting profit
- Net: they lose €5,000 relative to their best alternative
- **Correct decision: close the pizzeria and work as an employee**

Perfect Competition: The Conditions

Perfect competition: four conditions

1. **Many buyers and sellers:** no one is large enough to influence the price
2. **Homogeneous good:** the products of different firms are identical and perfect substitutes
3. **Perfect information:** everyone knows prices and quality
4. **Free entry and exit:** in the long run, new firms can enter without barriers

Perfect Competition: Implications

Consequence for the individual firm:

- The firm is a **price-taker**: it takes the price P^* as given by the market
- If it raises the price: it loses all its customers (they would buy from competitors)
- If it lowers the price: it simply earns less, with no new customers (it already had them all)

Examples of markets close to PC

Wheat, corn, soybeans (world agricultural markets); foreign currencies (forex); some commodities (oil, minerals); stock markets for liquid securities.

Revenue of the Price-Taking Firm (1)

For the price-taking firm the price P is constant.

Total revenue:

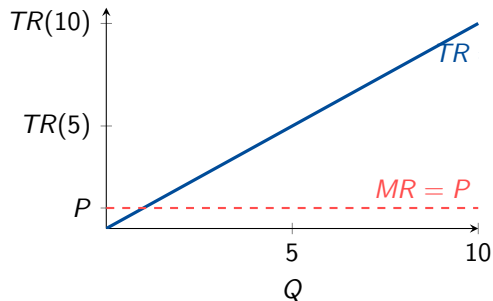
$$TR(Q) = P \cdot Q \quad (\text{straight line through the origin, slope } P)$$

Marginal revenue:

$$MR = \frac{dTR}{dQ} = P$$

The additional revenue from each unit produced is *exactly* equal to the price: I do not have to lower the price to sell one more unit (unlike monopoly).

Revenue of the Price-Taking Firm (2)



Properties:

- TR is a straight line with slope P
- $MR = P$: constant
- Unlike monopoly, where $MR < P$

The Maximization Problem (1)

The firm chooses $Q \geq 0$ to maximize:

$$\max_{Q \geq 0} \pi(Q) = TR(Q) - TC(Q) = P \cdot Q - C(Q)$$

First-Order Condition (FOC)

$$\frac{d\pi}{dQ} = \frac{dTR}{dQ} - \frac{dTC}{dQ} = MR - MC = P - MC(Q) = 0$$

$$\boxed{P = MC(Q^*)}$$

The Maximization Problem (2)

Second-Order Condition (SOC)

$$\frac{d^2\pi}{dQ^2} = -\frac{dMC}{dQ} < 0 \iff MC'(Q^*) > 0$$

MC must be **increasing** at the optimal point. If it is decreasing, we are at a *minimum* of profit, not a maximum.

Why $P = MC$?

The condition $P = MC$ is simply the principle of marginal analysis applied to production.

Unit-by-unit reasoning:

If $P > MC$

Producing one more unit adds P to revenue and MC to costs.
Since $P > MC$: profit **increases**.
⇒ It pays to **expand**.

If $P < MC$

The last unit produced costs more than it brings in.
Profit **decreases** by producing that unit.
⇒ It pays to **contract**.

If $P = MC$

Neither expanding nor contracting improves profit.
Profit is **stationary**.
If MC is increasing: it is a **maximum**.

Conclusion: the firm should keep producing as long as the additional revenue (P) exceeds the additional cost (MC). It stops when they are equal.

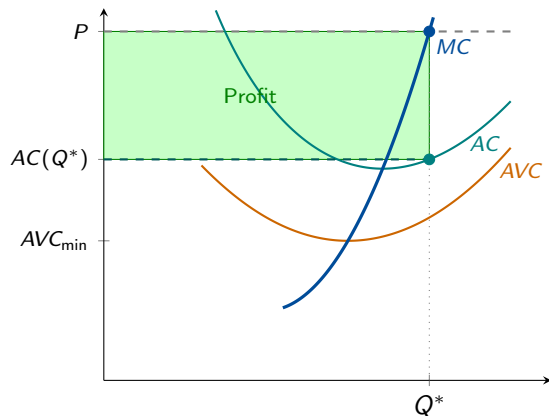
Example

Firm with $P = 12$ and marginal costs:

Unit	MC	P	$MR - MC$	Produce?
1	3	12	+9	Yes
2	5	12	+7	Yes
3	8	12	+4	Yes
4	11	12	+1	Yes
5	12	12	0	Indifferent ($Q^* = 5$)
6	14	12	-2	No
7	17	12	-5	No

Reading: the first 5 units are produced because each one adds profit. The 6th would cost 14 but bring in only 12: it pays to stop at 5.

Profit as an Area: Positive Profit Case



How to read the graph:

1. The firm chooses Q^* where: $MC = P$.
2. With $P = 15$, the optimum is: $Q^* = 4$.
3. At Q^* :

$$AC(Q^*) = 9.5.$$

4. The per-unit margin is:

$$P - AC(Q^*) = 15 - 9.5 = 5.5.$$

5. Total profit is the green rectangle:

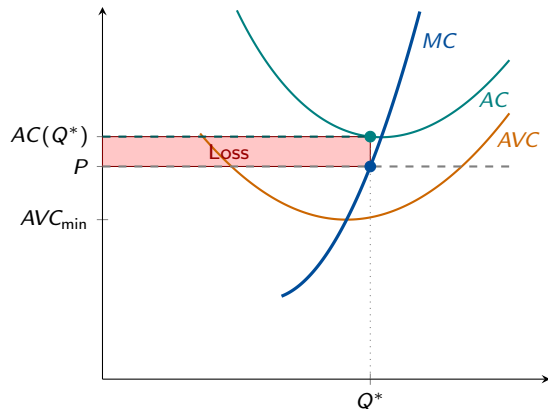
$$\pi = (15 - 9.5) \cdot 4 = 22.$$

Since

$$P > AC(Q^*),$$

the firm earns a **positive profit**.

Profit as an Area: Loss Case



How to read the graph:

1. The firm chooses Q^* where: $MC = P$.
2. With $P = 8$, the optimum is approximately: $Q^* \approx 3.291$

3. At Q^* :

$$AC(Q^*) \approx 9.123.$$

4. The per-unit margin is:

$$P - AC(Q^*) \approx 8 - 9.123 = -1.123.$$

5. Profit, which is negative, is represented by the red rectangle:

$$\pi = (8 - 9.123) \cdot 3.291 \approx -3.70.$$

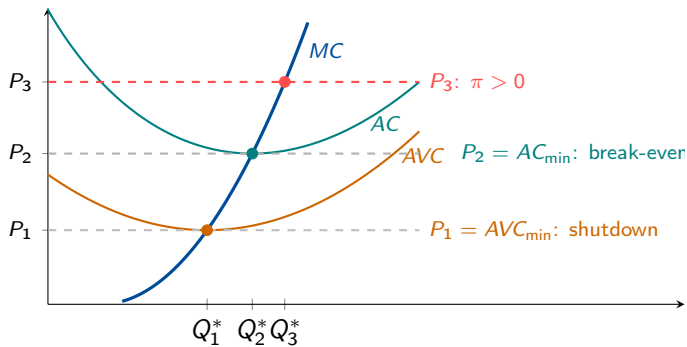
Since

$$P < AC(Q^*),$$

the firm incurs a **loss**.

However, it **produces in the short run** because $P = 8 > AVC_{\min} = 6$

Three Profit Cases: Complete Graph



- $P_1 = AVC_{\min} = 6$: shutdown point. The firm is indifferent between producing and not producing: in both cases $\pi = -FC$.
- $P_2 = AC_{\min} \approx 9.1$: break-even point. $\pi = 0$. In the long run the firm is indifferent between staying and exiting.
- $P_3 = 12 > AC_{\min}$: positive profit. The firm produces and earns $\pi > 0$.

When Is It Optimal Not to Produce?

The firm always has an alternative to production: **not producing** ($Q = 0$).

$$\pi(0) = P \cdot 0 - C(0) = -FC$$

The choice: produce ($Q^* > 0$) vs not produce ($Q = 0$).

The firm produces if: $\pi(Q^*) \geq \pi(0)$

$$PQ^* - C(Q^*) \geq -FC$$

$$PQ^* \geq C(Q^*) - FC = VC(Q^*)$$

$$P \geq \frac{VC(Q^*)}{Q^*} = AVC(Q^*)$$

Shutdown condition (short run)

The firm produces if and only if:

$$P \geq AVC_{\min}$$

The shutdown point is the **minimum of the AVC curve**.

Why Fixed Costs Do Not Matter for the Decision to Produce (1)

This is one of the most important (and counterintuitive) results in microeconomics.

Scenario: restaurant with $FC = €5,000/\text{month}$

A restaurant has already paid the €5,000 rent for the month. At the end of the month it faces the following scenarios:

- If it opens: revenue €8,000, variable costs €4,000 $\Rightarrow \pi = 8,000 - 4,000 - 5,000 = -1,000$
- If it stays closed: $\pi = -5,000$

What should the restaurant do?

$$\pi(\text{open}) = -1,000 > -5,000 = \pi(\text{closed})$$

$\Rightarrow$ It pays to **open**, even though it makes a loss!

Why Fixed Costs Do Not Matter for the Decision to Produce (2)

Why? The rent has already been paid (sunk cost)—it does not change with the decision to open or not. By opening, the restaurant covers the €4,000 of variable costs with the €8,000 of revenue and recovers €4,000 of the rent. The overall loss is €1,000 instead of €5,000.

General rule

In the short run, fixed costs are **sunk** (unrecoverable). They do not affect the decision to produce: all that matters is whether revenue covers **variable costs**.

Shutdown in the SR vs. LR (1)

Short run

Fixed costs are **sunk**: already incurred, not recoverable.

The decision to produce depends only on variable costs.

Produces if: $P \geq AVC_{\min}$

Even with $\pi < 0$: it pays to produce if losses are smaller than FC .

Long run

All costs become variable (contracts expire, plants can be dismantled).

$FC = 0$: there are no sunk costs.

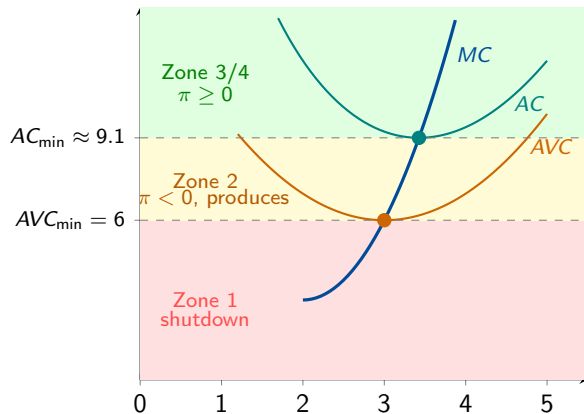
Produces if: $P \geq AC_{\min}$

If $\pi < 0$ in the LR: the firm exits the market.

Shutdown in the SR vs. LR (2)

Zone	Condition	SR decision	LR decision
1	$P < AVC_{\min}$	Does not produce	Exits
2	$AVC_{\min} \leq P < AC_{\min}$	Produces ($\pi < 0$)	Exits
3	$P = AC_{\min}$	Produces ($\pi = 0$)	Indifferent
4	$P > AC_{\min}$	Produces ($\pi > 0$)	Stays

The Four Zones



The **SR supply curve** is the portion of MC in zones 2, 3, 4:

$$Q^s(P) = \begin{cases} Q : MC(Q) = P & P \geq AVC_{\min} \\ 0 & P < AVC_{\min} \end{cases}$$

Shutdown point = minimum of AVC

Break-even point = minimum of AC

The supply curve is **increasing**: higher $P \Rightarrow$ higher Q^* .

Exercise 1

Giulia owns a clothing store. Annual data:

- Revenue: €180,000
- Rent, suppliers, employees: €110,000
- Giulia's alternative salary (manager at another company): €52,000
- Capital invested in the business: €80,000, alternative return 3%

1. Compute accounting profit and economic profit.
2. Is it worthwhile for Giulia to continue the business?
3. If revenue fell to €160,000, what would change?

Solution

1. Opportunity cost of capital: $80,000 \times 3\% = 2,400$.

$$\pi_{\text{acc}} = 180,000 - 110,000 = 70,000$$

$$\pi_{\text{econ}} = 180,000 - 110,000 - 52,000 - 2,400 = 15,600$$

2. $\pi_{\text{econ}} = +15,600 > 0$: Giulia earns €15,600 more than her best alternative (salary + return on capital). **It pays to continue.**

Beware of a common confusion

It is not enough that $\pi_{\text{acc}} > 0$. Giulia gives up €52,000 in salary + €2,400 in return = €54,400 of implicit costs. With accounting profit of €70,000, the net advantage is only €15,600.

Solution

3. Revenue falls to €160,000:

$$\pi_{\text{acc}} = 160,000 - 110,000 = 50,000 > 0$$

$$\pi_{\text{econ}} = 160,000 - 110,000 - 52,000 - 2,400 = -4,400 < 0$$

Interpretation:

- Accounting profit is still positive (€50,000)
- Economic profit is negative (−€4,400): Giulia earns less than she would get by closing the store and working as a manager

Decision

In the **short run**: it depends on fixed costs (if there are ongoing contracts, it will pay to finish the period). In the **long run**: Giulia should close the store and work as a manager. The economically correct choice is always based on *economic* profit, not accounting profit.

Exercise 2

A price-taking firm has $C(Q) = Q^3 - 6Q^2 + 15Q + 10$. Market price $P = 9$.

1. Derive MC , AVC and find $AVC_{\min}$.
2. Find Q^* with the FOC. Check the SOC.
3. Compute profit. Does the firm produce in the SR?

Solution

1. $FC = 10$, $VC(Q) = Q^3 - 6Q^2 + 15Q$.

$$MC = C'(Q) = 3Q^2 - 12Q + 15$$

$$AVC = \frac{VC}{Q} = Q^2 - 6Q + 15$$

Minimum of AVC:

$$\frac{d(AVC)}{dQ} = 2Q - 6 = 0 \Rightarrow Q = 3$$

$$AVC_{\min} = AVC(3) = 9 - 18 + 15 = 6$$

Check: $MC(3) = 27 - 36 + 15 = 6 = AVC_{\min}$ ✓

MC intersects AVC at its minimum, as expected.

Implication

The firm produces (in the SR) only if $P \geq 6$. With $P = 9 > 6$: yes, it produces.

Solution

2. FOC: $P = MC$

$$9 = 3Q^2 - 12Q + 15 \Rightarrow 3Q^2 - 12Q + 6 = 0 \Rightarrow Q^2 - 4Q + 2 = 0$$

$$Q = \frac{4 \pm \sqrt{16 - 8}}{2} = 2 \pm \sqrt{2}$$

Two solutions: $Q_1 = 2 - \sqrt{2} \approx 0.59$ and $Q_2 = 2 + \sqrt{2} \approx 3.41$.

SOC: $MC'(Q) = 6Q - 12$. It must be > 0 .

- $Q_1 \approx 0.59 < 2$: $MC' = 6(0.59) - 12 < 0 \Rightarrow MC$ decreasing $\Rightarrow$ profit **minimum**. Discarded.
- $Q_2 \approx 3.41 > 2$: $MC' = 6(3.41) - 12 > 0 \Rightarrow MC$ increasing $\Rightarrow$ **maximum**. ✓

$$Q^* = 2 + \sqrt{2} \approx 3.41$$

Solution

3. Numerical computation at $Q^* = 2 + \sqrt{2} \approx 3.414$:

$$TR = 9 \times 3.414 = 30.73$$

$$C(Q^*) = (3.414)^3 - 6(3.414)^2 + 15(3.414) + 10 \approx 39.80 - 69.94 + 51.21 + 10 = 31.07$$

$$\pi = 30.73 - 31.07 \approx -0.34 \quad (= 4\sqrt{2} - 6)$$

Shutdown check: $AVC(Q^*) = (3.41)^2 - 6(3.41) + 15 \approx 11.63 - 20.46 + 15 = 6.17$
 $P = 9 > 6.17 = AVC(Q^*)$ ✓ **The firm produces.**

Interpretation

The firm makes a small loss ($\pi \approx -0.34$), but it produces anyway because:

- Loss when producing: ≈ -0.34
- Loss when not producing: $-FC = -10$

Producing is **much better** than staying idle.

Exercise 3

Same firm: $C(Q) = Q^3 - 6Q^2 + 15Q + 10$, $AVC_{\min} = 6$, $AC_{\min} \approx 9.1$. For prices $P \in \{4, 6, 9, 12\}$:

1. Find Q^* (or indicate shutdown).
2. Compute π .
3. State the SR and LR decision.

Solution

Functions: $MC = 3Q^2 - 12Q + 15$, $AVC = Q^2 - 6Q + 15$, $AC = Q^2 - 6Q + 15 + 10/Q$

P	Q^*	$AVC(Q^*)$	$AC(Q^*)$	π	SR / LR decision
4	—	—	—	-10	Shutdown / Exits
6	3.00	6.00	9.33	-10.00	Indifferent / Exits
9	3.41	6.17	9.10	-0.34	Produces / Exits
12	3.73	6.54	9.21	+10.39	Produces / Stays

Notes:

- $P = 4 < AVC_{\min} = 6$: shutdown. Loss = $-FC = -10$.
- $P = 6 = AVC_{\min}$: exact shutdown point. $\pi = -FC = -10$: identical to not producing.
- $P = 9$: small loss, but it produces (loss $-0.34 > -FC = -10$). LR: exits because $P < AC_{\min}$.
- $P = 12 > AC_{\min}$: positive profit. In the LR new firms enter.

Exercise 4

A firm has $C(Q) = Q^2 + 4Q + 25$ (with $FC = 25$).

1. Find MC , AVC , $AVC_{\min}$, $AC_{\min}$.
2. At what price is the firm at the LR break-even point ($\pi = 0$)?
3. Compute Q^* and π for $P = 8$, $P = 10$, $P = 14$.
4. For each price, state the SR and LR decision.

Solution

1.

$$MC = 2Q + 4, \quad AVC = \frac{Q^2 + 4Q}{Q} = Q + 4$$

AVC is increasing ($dAVC/dQ = 1 > 0$): $AVC_{\min} = \lim_{Q \rightarrow 0^+} AVC = 4$ (intercept).

$AC = Q + 4 + 25/Q$. $dAC/dQ = 1 - 25/Q^2 = 0 \Rightarrow Q = 5$.

$$AC_{\min} = AC(5) = 5 + 4 + 5 = 14$$

2. The LR break-even point is $P = AC_{\min} = 14$.

Check: $P = MC \Rightarrow 14 = 2Q + 4 \Rightarrow Q^* = 5$.

$\pi = 14 \cdot 5 - C(5) = 70 - (25 + 20 + 25) = 70 - 70 = 0 \checkmark$

Solution

3. and 4. FOC: $P = MC = 2Q + 4 \Rightarrow Q^* = (P - 4)/2$.

P	Q^*	$C(Q^*)$	π	SR / LR decision
8	2	$4 + 8 + 25 = 37$	$16 - 37 = -21$	Produces / Exits ($P < AC_{\min} = 14$)
10	3	$9 + 12 + 25 = 46$	$30 - 46 = -16$	Produces / Exits
14	5	$25 + 20 + 25 = 70$	$70 - 70 = 0$	Produces / Indifferent

- $P = 8 > AVC_{\min} = 4$: produces in the SR even though $\pi < 0$. In the LR it exits ($P < AC_{\min} = 14$).
- $P = 10 > AVC_{\min} = 4$: same logic. In the LR it exits.
- $P = 14 = AC_{\min}$: exact break-even. In the LR indifferent (stays if the others exit).

Note: With $P = 8$ the firm produces $Q^* = 2$ and loses €21. But by not producing it would lose €25 (just the FC). Producing reduces the loss by €4.

General Summary

- Economic profit: $\pi = PQ - C(Q)$, includes implicit costs
- $\pi = 0$: normal return, not the absence of earnings
- Price-taker optimum: $P = MC(Q^*)$ with $MC'(Q^*) > 0$
- Profit as an area: $\pi = (P - AC(Q^*)) \cdot Q^*$
- SR shutdown: produces if $P \geq AVC_{\min}$ (FC sunk)
- LR shutdown: produces if $P \geq AC_{\min}$

Skills Acquired

- ✓ Distinguish accounting and economic profit
- ✓ Set up and solve $\max_Q \pi(Q) = PQ - C(Q)$
- ✓ Apply the FOC and SOC and select the maximum
- ✓ Represent and compute profit as an area
- ✓ Apply the shutdown condition in the SR and LR
- ✓ Classify the firm's situation into the four zones

Optional Homework Exercises

1. An entrepreneur has revenue of €200,000, explicit costs of €130,000, an alternative salary of €55,000, and invested capital of €100,000 at 4%. Compute π_{acc} and π_{econ} . Is it worthwhile to continue?
2. $C(Q) = 2Q^2 + 8Q + 32$. Find MC , $AVC_{\min}$, $AC_{\min}$. For $P = 14$: find Q^* , π , SR/LR decision. For $P = 24$: same analysis.
3. $C(Q) = Q^3 - 4Q^2 + 6Q + 3$. Find $AVC_{\min}$ and $AC_{\min}$. For which values of P does the firm: (a) not produce; (b) produce but make a loss; (c) make a positive profit?
4. True or false (explain):
 - a) A firm with $\pi_{\text{acc}} > 0$ should always stay open.
 - b) A firm that makes losses must close immediately.
 - c) Fixed costs affect the decision on how many units to produce.
5. A firm has $FC = 200$, $AVC_{\min} = 15$, $AC_{\min} = 25$. The price is $P = 20$. Describe the situation (zone, SR/LR decision) and compute the SR loss if $Q^* = 10$ and $AC(10) = 36$.